PROJECT SYNOPSIS

DESIGN AND WORKING OF A VACUUM CLEANER

Electrical & Electronics Engineering (EEE)

Team: Clean Tech Innovators



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A project submitted for academic exhibition

Project Overview

This project presents a simple vacuum cleaner model that demonstrates the use of a motor-driven fan to create airflow and suction. The model helps explain the pressure difference that draws dust into a collection chamber.

Submitted by: Clean Tech Innovators

1. INTRODUCTION

A vacuum cleaner is an electrical cleaning device used to remove dust, dirt and small particles from floors, furniture and other surfaces. It draws air through an inlet nozzle with the help of a motor-driven fan. The dust carried by the incoming air is collected in a container or dust chamber, while the air is allowed to pass through an outlet.

This project demonstrates the basic operation of a vacuum cleaner using simple components. It helps us understand the practical use of an electric motor, airflow, suction and the conversion of electrical energy into mechanical energy. The model is designed for educational demonstration and may differ in construction and performance from a commercial vacuum cleaner.

2. OBJECTIVES

• To study the basic construction and working of a vacuum cleaner.

• To understand the principle of suction caused by a pressure difference.

• To demonstrate the use of a motor and fan/impeller for producing airflow.

• To observe how dust particles can be drawn into a collection chamber.

• To relate electrical energy, mechanical motion and movement of air in an everyday appliance.

3. WORKING PRINCIPLE

The vacuum cleaner works on the principle of pressure difference and suction. When the motor rotates the fan or impeller, air is moved from the inside of the cleaner towards the outlet. This produces a lower-pressure region near the inlet compared with the surrounding atmospheric pressure.

Because of this pressure difference, atmospheric pressure pushes air and loose dust particles through the suction nozzle and into the cleaner. The dust is retained in the collection chamber or filter, depending on the model, while the air passes through the air path. Therefore, suction is produced by the movement of air and the pressure difference—not by the cleaner creating a perfect vacuum.

4. MAIN COMPONENTS AND THEIR FUNCTIONS

1. DC Motor: Converts electrical energy into rotational mechanical energy and drives the fan.
2. Fan/Impeller: Moves air and helps create the pressure difference required for suction.
3. Power Supply/Battery: Supplies electrical energy to the motor.
4. Switch: Turns the model ON and OFF.
5. Suction Pipe/Nozzle: Directs dust and air from the surface towards the collection chamber.
6. Dust Collection Container: Collects the dust particles drawn into the model.
7. Connecting Wires: Carry electrical current between the supply, switch and motor.
8. Filter (if fitted): Helps prevent collected dust from travelling with the outgoing air.

5. CONSTRUCTION OF THE MODEL

The model is assembled by fitting the motor and fan securely in the body of the cleaner. The motor is connected to the power supply through a switch using suitable connecting wires. A suction pipe or nozzle is attached to the inlet side, and a container is provided to collect dust. If a filter is included in the model, it is placed so that dust can be retained while air continues to flow.

All parts should be fixed firmly, and the air path should be kept as clear as possible. The arrangement of parts depends on the actual model prepared by the team. The model should be operated only with the intended low-voltage supply and under the supervision of a teacher or responsible adult.

6. WORKING PROCEDURE

• Step 1: Check that the motor, fan, switch, wires, nozzle and collection chamber are properly fitted.

• Step 2: Connect the model to its specified power supply or battery.

• Step 3: Turn the switch ON. Electrical energy is supplied to the motor.

• Step 4: The motor rotates the fan/impeller, which moves air through the cleaner.

• Step 5: The pressure difference causes air and loose dust to enter through the suction nozzle.

• Step 6: Dust particles collect in the dust chamber or filter, while air passes through the outlet.

• Step 7: Turn the switch OFF after the demonstration and disconnect the supply before inspecting or cleaning the model.

7. ENERGY CONVERSION

The main energy conversion in the model is

ELECTRICAL ENERGY → MECHANICAL ENERGY → MOVEMENT OF AIR

Electrical energy operates the motor. The motor produces rotational mechanical motion, and the fan converts this motion into airflow. The resulting pressure difference creates suction.

8. APPLICATIONS

• Cleaning dust from floors and small surfaces.

• Removing loose dust from corners and hard-to-reach places.

• Cleaning furniture and workspaces, where suitable.

• Demonstrating motor operation, airflow and pressure difference in educational laboratories.

• Understanding the basic design of domestic and portable cleaning appliances.

9. ADVANTAGES AND LIMITATIONS

Advantages

• Easy to demonstrate and operate.

• Helps explain a practical application of an electric motor.

• Can save time and effort compared with collecting loose dust manually.

• A small model can be portable, depending on its design.

Limitations

• Suction depends on the motor, fan design, air leaks and power supply.

• A simple educational model may not remove heavy dirt or fine dust effectively.

• A suitable dust container and filter are important for keeping dust from escaping.

10. SAFETY PRECAUTIONS

• Use only the power supply specified for the model.

• Keep hands, loose clothing and objects away from the rotating fan.

• Do not operate the model with damaged wires or loose connections.

• Keep liquids away from the motor and electrical connections.

• Switch OFF and disconnect the supply before cleaning the container or checking components.

• Operate the model under teacher/adult supervision.

11. CONCLUSION

The vacuum cleaner model demonstrates how a motor-driven fan produces airflow and a pressure difference to draw dust into a collection chamber. It shows the conversion of electrical energy into mechanical energy and then into the movement of air. Through this project, we gained practical understanding of basic electrical components, suction, airflow and the working of a commonly used household appliance.

12. REFERENCES

1. Basic Electrical Engineering and Electrical Workshop classroom notes.

2. Instructional material and demonstrations related to electric motors and airflow.

3. Observation and testing of the team’s vacuum cleaner model.